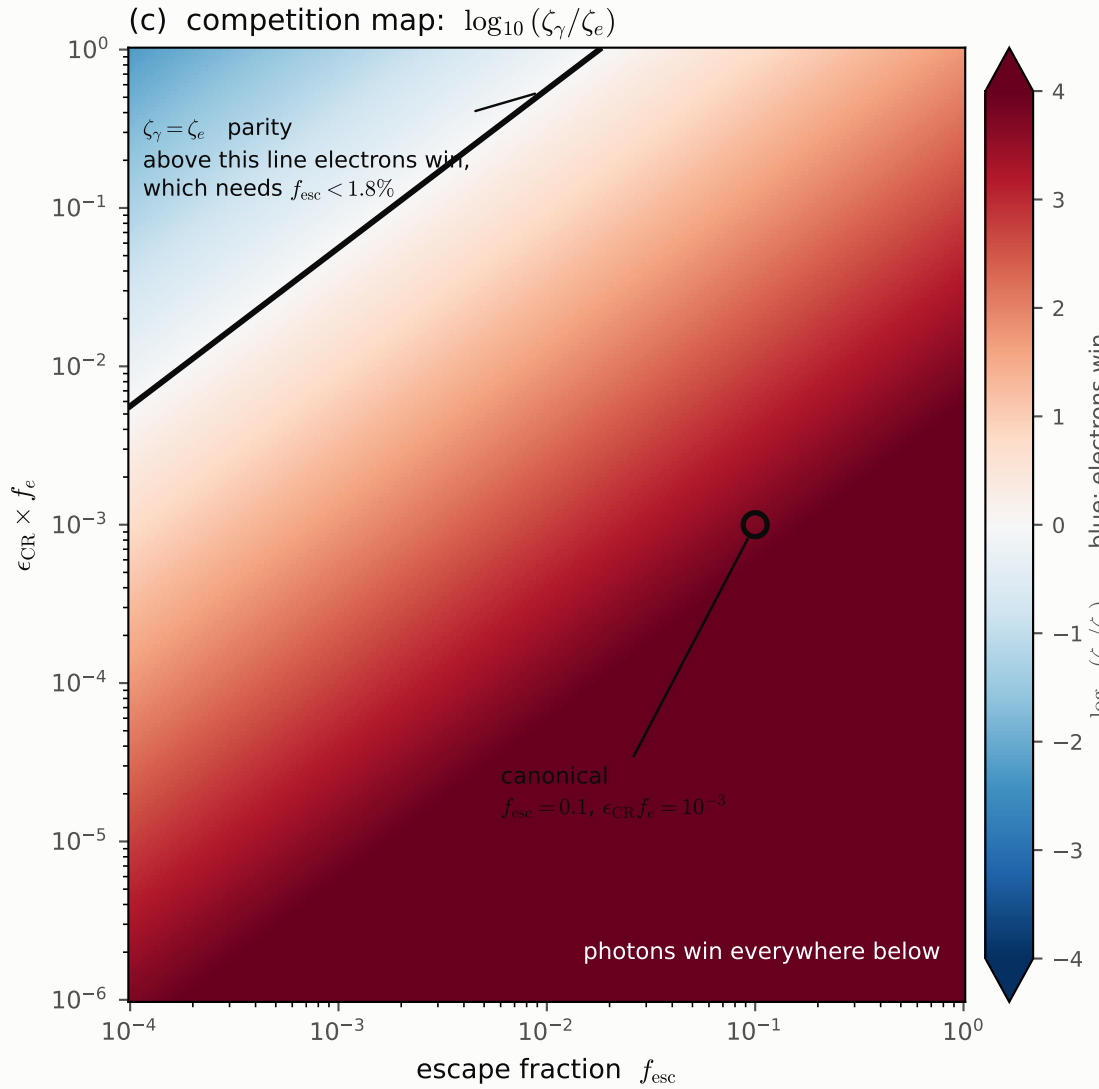
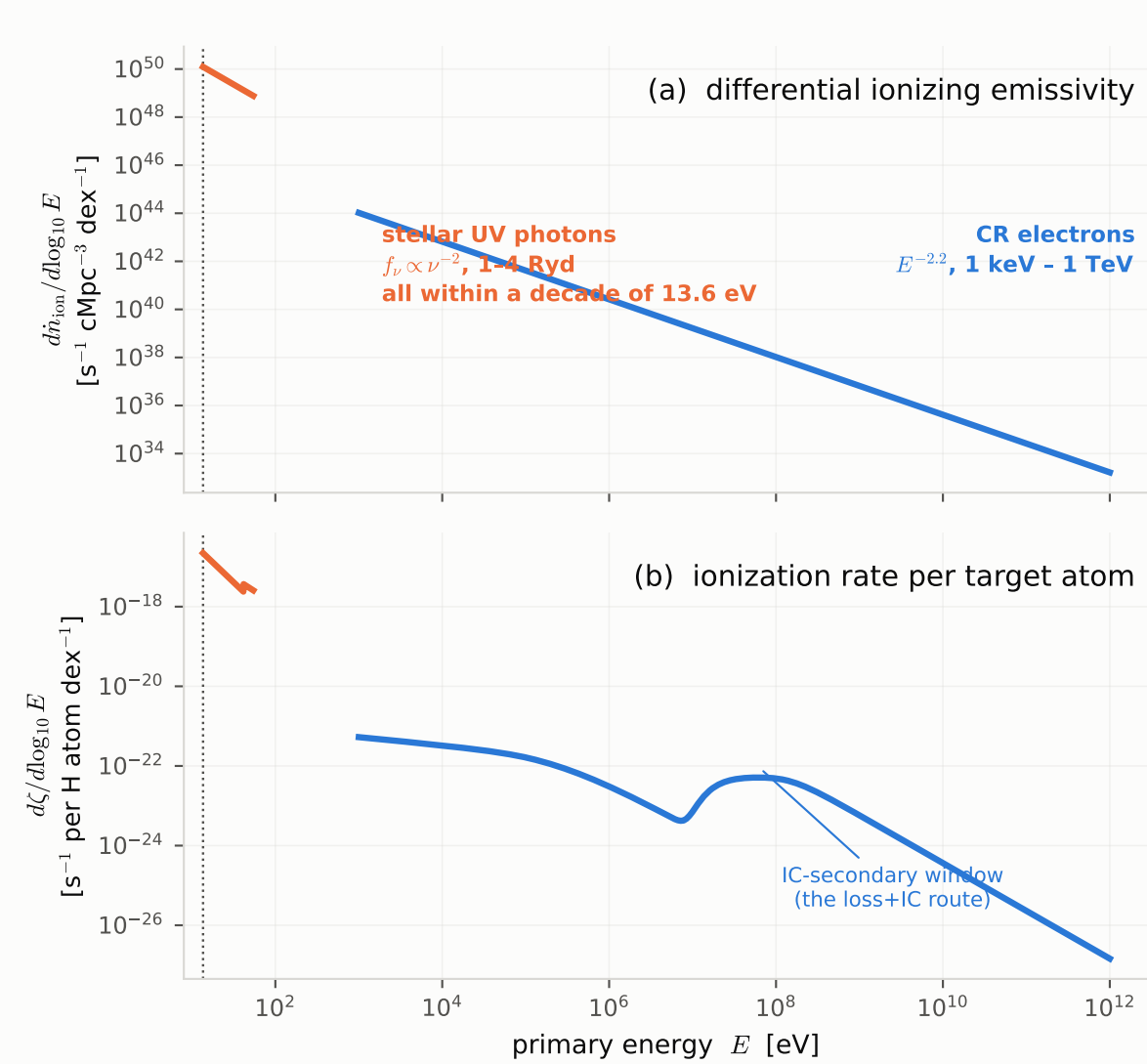


Stellar UV photons versus cosmic-ray electrons as reionization agents at $z = 10$



IGM at $z = 10$, $x_e = 10^{-4}$, pure H.
Electron yields: the loss+IC route — igm_losses.py: 7 loss mechanisms, RBED event counting, BED secondary cascade, Klein-Nishina IC, + IC-secondary photoionization.
 ρ_{UV} : Donnan+24 JWST PRIMER. $\xi_{\text{ion}} = 10^{25.28}$: Llerena+25. $\rho_{\text{UV}} \rightarrow \rho_{\text{SFR}}$: Madau & Dickinson 14 (Salpeter).
SN/ $M_\odot$ derived from the IMF, not adopted.
Stellar SED: population of U37126 ($z = 10.255$, Marques-Chaves+26, BPASS v2.2.1, $Z = 0.003$, 6.8 Myr); LyC shape parametrised, α bracketed 1-3.
Photons lead by 3.7 decades at canonical parameters; parity would need $\epsilon_{\text{CR}} f_e = 5.6$, which exceeds unity.